

4200-SCS



- **Intuitive, point-and-click Windows®-based environment**
- **Unique Remote PreAmps extend the resolution of SMUs to 0.1fA**
- **C-V instrument makes C-V measurements as easy as DC I-V**
- **Pulse and pulse I-V capabilities for advanced semiconductor testing**
- **Scope card provides integrated scope and pulse measure functionality**
- **Self-contained PC provides fast test setup, powerful data analysis, graphing and printing, and on-board mass storage of test results**
- **Unique browser-style Project Navigator organizes tests by device type, allows access to multiple tests, and provides test sequencing and looping control**
- **Built-in stress/measure, looping, and data analysis for point-and-click reliability testing, including five JEDEC-compliant sample tests**
- **Integrated support for a variety of LCR meters, Keithley switch matrix configurations, and both Keithley Series 3400 and Agilent 81110 pulse generators**
- **Includes software drivers for leading analytical probes**

Semiconductor Characterization System

DC I-V, C-V, and Pulse in One Test Environment

The easy-to-use Model 4200-SCS Semiconductor Characterization System performs lab grade DC I-V, C-V, and pulse device characterization, real-time plotting, and analysis with high precision and sub-femtoamp resolution. The 4200-SCS offers the most advanced capabilities available in a fully integrated characterization system, including a complete, embedded PC with Windows operating system and mass storage. Its self-documenting, point-and-click interface speeds and simplifies the process of taking data, so users can begin analyzing their results sooner. Additional features enable stress-measure capabilities suitable for a variety of reliability tests.

The powerful test library management tools included allow standardizing test methods and extractions to ensure consistent test results. The Model 4200-SCS offers tremendous flexibility with hardware options that include four different switch matrix configurations and a variety of LCR meters and pulse generators. Customer support packages are also available, including applications support, calibration, repair, and training.

A Total System Solution

The Model 4200-SCS provides a total system solution for DC I-V, C-V, and pulse characterization and reliability testing of semiconductor devices, test structures, and materials. This advanced parameter analyzer provides intuitive and sophisticated capabilities for a wide variety of semiconductor tests. The Model 4200-SCS combines unprecedented measurement speed and accuracy with an embedded Windows-based PC and the Keithley Interactive Test Environment (KITE) to provide a powerful single-box solution. KITE allows users to gain familiarity quickly with tasks such as managing tests and results and generating reports. Sophisticated and simple test sequencing and external instrument drivers simplify performing automated device and wafer testing with combined I-V, C-V, and pulse measurements. The exceptional low current performance of the Model 4200-SCS makes it the perfect solution for research studies of single electron transistors (SETs), molecular electronic devices, and other nanoelectronic devices that require I-V characterization. The Model 4200-SCS can be used to make four-probe van der Pauw resistivity and Hall voltage measurements, eliminating the need for a switch matrix and user-written code. With remote preamps added, resistances well above $10^{12}\Omega$ can be measured.

The Model 4200-SCS is modular and configurable. The system supports up to nine Source-Measure Units (SMUs) in any combination of medium and high power SMUs. A high-power SMU provides 1A/20W capability. Also available are the C-V option and the pulse and scope pulse measure modules. The C-V option includes the C-V Power package, which supports high power C-V measurements up to 400V and 300mA, up to 60V of differential DC bias, and quasistatic C-V measurements.

Applications Packages

By combining specific sets of hardware with Keithley-developed code and interconnect, a variety of application packages are offered that expand the Model 4200-SCS's pulsed testing capabilities. The 4200-PIV-A package performs charge trapping and isothermal testing for leading-edge CMOS research. The 4200-PIV-Q package is designed for higher power pulse testing in III-V, LDMOS, and other higher frequency and higher power FET devices. The 4200-FLASH package tests floating gate FLASH and embedded NVM memory.

Extended Measurement Resolution

An optional Remote PreAmp, the Model 4200-PA, extends the system's measurement resolution from 100fA to 0.1fA by effectively adding five current ranges to either SMU model. The PreAmp module is fully integrated with the system; to the user, the SMU simply appears to have additional measurement resolution available. The Remote PreAmp is shipped installed on the back panel of the Model 4200-SCS for local operation. This installation allows for standard cabling to a probe, test fixture, or switch matrix. Users can remove the PreAmp from the back panel and place it in a remote location (such as in a light-tight enclosure or on the probe platen) to eliminate measurement problems due to long cables. Platen mounts and triax panel mount accessories are available.

KTE Interactive Software Tools

KTE Interactive includes four software tools for operating and maintaining the Model 4200-SCS in addition to the Windows operating system:

- The Keithley Interactive Test Environment (KITE) is the Model 4200-SCS Windows device characterization application. It provides advanced test definition, parameter analysis and graphing, and automation capabilities required for modern semiconductor characterization. Built-in looping, stress-measure capabilities, and data management enable many types of reliability testing.

4200-SCS

Ordering Information

4200-SCS/F

Flat Panel Display

4200-SCS/C

Composite Front Bezel; requires an external SVGA display

Accessories Supplied

Reference and User Manual on CD-ROM

236-ILC-3 Interlock Cable, 3m

Note: All 4200-SCS systems and instrument options are supplied with required cables of 2m length.

Additional Instrumentation

4210-CVU

Integrated C-V Instrument

4205-PG2

Dual-Channel Pulse Generator

4200-SCP2

Dual-Channel Digital Oscilloscope

4200-SCP2HR

High Resolution, Dual Channel Integrated Oscilloscope

4200-PIV-A

Complete Pulse I-V Package for leading edge CMOS

4200-PIV-Q

Pulse I-V Package with Q point and dual-channel pulsing

4200-FLASH

Non-volatile Memory Test Package

4200-SCP2-ACC

Optional Scope Probe

Related Products

707A Semiconductor Switching Matrix Mainframe

708A Single Slot Switching Matrix Mainframe

4200-SCP2-ACC
70MHz Scope Probe

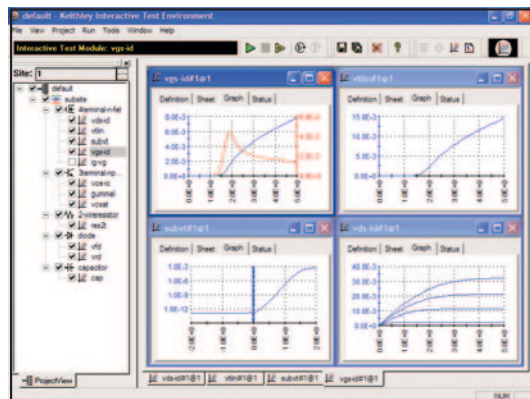
7072 8x12 Semiconductor Matrix Card

7072-HV 8x12 High Voltage Semiconductor Matrix Card

7174A 8x12 High Speed, Low Current Matrix

Semiconductor Characterization System

DC I-V, C-V, and Pulse in One Test Environment



The Keithley Interactive Test Environment (KITE) is designed to let users understand device behavior quickly. When running a test sequence, users can view results and plots for completed tests while the sequence is still running. As shown here, multiple plots can be viewed at the same time to get a complete picture of device performance.

- Keithley User Library Tool (KULT)—Allows test engineers to integrate custom algorithms into KITE using Model 4200-SCS or external instruments.
- Keithley Configuration Utility (KCON)—Allows test engineers to define the configuration of GPIB instruments, switch matrices, and analytical probes connected to the Model 4200-SCS. It also provides system diagnostics functions.
- Keithley External Control Interface (KXCI)—The Model 4200-SCS application for controlling the Model 4200-SCS from an external computer via the GPIB bus.

KITE Projects

A project is a collection of related tests, organized in a hierarchy that parallels the physical layout of the devices on a wafer. KITE operates on projects using an interface called the project navigator. The project navigator simplifies organizing test files, test execution, and test sequencing. The project navigator organizes tests into a logical hierarchy presented in a browser style format. This structure allows users to define projects around wafer testing:

- The project level organizes subsites and controls wafer looping execution.
- The subsite level organizes devices and controls subsite test sequencing.
- The device level organizes test modules, manages test module libraries, and controls device test sequencing.
- The test module level performs tests, analyzes data, and plots results.

Prober Control

Keithley provides integrated prober control for supported analytical probes when test sequencing is executed on a user-programmable number of probe sites on a wafer. Contact the factory for a list of supported analytical probes. A manual prober mode prompts the operator to perform prober operations during the test sequence.

Test Sequencing

KITE provides "point and click" test sequencing on a device, a group of devices (subsite, module, or test element group), or a user-programmable number of probe sites on a wafer. One sequence can include DC I-V, C-V, and pulse tests.

Keithley User Library Tool (KULT)

The Keithley User Library Tool is an open environment that provides you with the flexibility to create your own custom routines as well as use existing Keithley and third-party C-language subroutine libraries. User library modules are accessed in KITE through User Test Modules. Factory supplied libraries provide up and running capability for supported instruments. Users can edit and compile subroutines, then integrate libraries of subroutines with KITE, allowing the Model 4200-SCS to control an entire test rack from a single user interface. KULT is derived from the Keithley S600 and Series S400 Parametric Test Systems. This simplifies migration of test libraries between the Model 4200-SCS and Keithley parametric test systems.

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Semiconductor Characterization System DC I-V, C-V, and Pulse in One Test Environment

4210-CVU C-V Instrument

C-V measurements are as easy to perform as I-V measurements with the integrated C-V instrument. This optional capacitance-voltage instrument performs capacitance measurements from femtoFarads (fF) to nanoFarads (nF) at frequencies from 1kHz to 10MHz. The C-V option includes a new Power package that supports:

- High power C-V measurements up to 400V (200V per device terminal)—for testing high power devices, such as MEMs, LDMOS devices, displays, etc.
- DC currents up to 300mA—for measuring capacitance when a transistor is on.

The innovative design of the 4200-SCS has eight patents pending and is complemented by the broadest C-V test and analysis library available in any commercial C-V measurement solution. It also supplies diagnostic tools that ensure the validity of your C-V test results.

With this system, you can configure linear or custom C-V and C-f sweeps with up to 4096 data points. In addition, through the open environment of the 4200-SCS, you can modify any of the included tests, such as:

- C-V, C-t, and C-f measurements and analysis of:
 - New! Complete solar cell libraries, including DLCP
 - High and low κ structures
 - MOSFETs
 - BJTs
 - Diodes
 - III-V compound devices
 - Carbon nanotube (CNT) devices
- Doping profiles, T_{OX} , and carrier lifetime tests
- Junction, pin-to-pin, and interconnect capacitance measurements

The C-V instrument integrates directly into the Model 4200-SCS chassis. It can be purchased as an upgrade to existing systems or as an option for new systems.

4210-CVU: Selected C-V Specifications

MEASUREMENT FUNCTIONS

MEASUREMENT PARAMETERS: C_p -G, C_p -D, C_s -Rs, C_s -D, R-jX, Z-theta.
RANGING: Auto and fixed.

TEST SIGNAL

FREQUENCY RANGE: 1kHz to 10MHz.
SOURCE FREQUENCY ACCURACY: $\pm 0.1\%$.
SIGNAL OUTPUT LEVEL RANGE: 10mV rms to 100mV rms.
RESOLUTION: 1mV rms.
ACCURACY: $\pm(10.0\% + 1\text{mV rms})$ unloaded (at rear panel).

DC BIAS FUNCTION

DC VOLTAGE BIAS RANGE: $\pm 30\text{V}$ on both C-V HI and C-V LO ($\pm 60\text{V}$ differential).
RESOLUTION: 1.0mV.
ACCURACY: $\pm(0.5\% + 5.0\text{mV})$ unloaded.
MAXIMUM DC CURRENT: 10mA.

SWEEP CHARACTERISTICS

AVAILABLE SWEEP PARAMETERS: DC bias voltage, frequency, AC drive level.
SWEEP TYPE: Linear, Custom.
SWEEP DIRECTION: Up sweep, Down sweep.
NUMBER OF MEASUREMENT POINTS: 4096 points.

C-V POWER PACKAGE

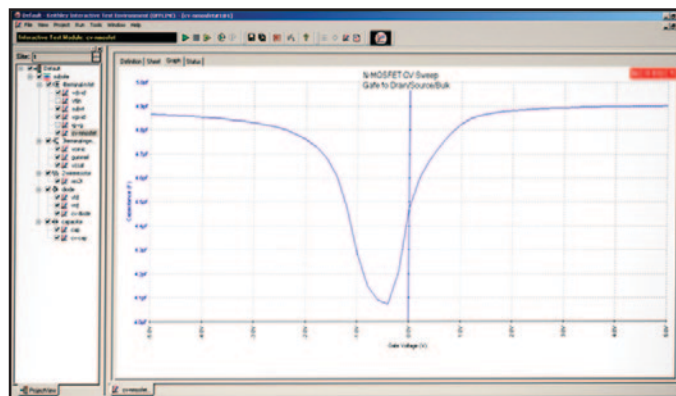
TYPICAL PERFORMANCE CHARACTERISTICS

MEASUREMENT PARAMETERS: C_p -Gp, DCV, timestamp.
RANGING: 1pF to 1nF.
MEASUREMENT TERMINALS: 2-wire SMA, with BNC adapters.
TEST SIGNAL: 100kHz to 10MHz, 10mV to 100mV.
DC VOLTAGE SOURCE: $\pm 200\text{V}$ with 5mV resolution ($\pm 400\text{V}$ differential).
DC CURRENT: 100mA or 300mA maximum.
TYPICAL CP ACCURACY @ 1MHz: 1.0%.
DC CURRENT SENSITIVITY: 10nA/V.
SMU BIAS TERMINALS SUPPORTED: 4.

RAMP RATE QUASISTATIC C-V

TYPICAL PERFORMANCE CHARACTERISTICS

MEASUREMENT PARAMETERS: C_p , DCV, timestamp.
RANGING: 1pF to 1nF.
MEASUREMENT TERMINALS: Triaxial guarded.
RAMP RATE: 0.1V/s to 1V/s.
DC VOLTAGE: $\pm 200\text{V}$.
TYPICAL CP ACCURACY: 5% at 1V/s ramp rate.



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Semiconductor Characterization System DC I-V, C-V, and Pulse in One Test Environment

Dual-Channel Pulse Generator

The optional, integrated dual-channel pulse generator adds pulsing to the Model 4200-SCS's DC source and measure capabilities. It supports voltage pulses as short as 10ns or up to $\pm 20V$ (into 50 Ω). Two pulse generators on one card provides you with the flexibility to apply pulses to two points on a DUT, such as the gate and the drain, simultaneously. The 4200-SCS can support up to four synchronized cards per system for eight pulse channels.

The 4205-PG2 supports two waveform generation modes in addition to the standard pulse mode. The Arbitrary Waveform mode can generate complex waveforms made up of up to 256K data points at clock speeds up to 25MHz. The Segment ARB™ mode (patent pending) simplifies creating, storing, and generating complex waveforms made from up to 1024 user-defined line segments. Each segment can have a different duration, allowing exceptional waveform generation flexibility.

Using a supplied User Test Module, it is simple to incorporate pulse generation into KITE test sequences. The pulse generator can also be used as a stand-alone pulse generator using the pulse generator's Window's GUI. This GUI can control a wide range of variables, including pulse frequency, duty cycle, rise/fall time, amplitude, offset, and the ability to trigger single pulses and/or pulse chains.

The dual-channel pulse generator has a wide range of uses. Typical applications include:

- Charge pumping to characterize interface state densities in MOSFET devices
- Using AC stress pulses of varying frequencies to simulate real-world AC signals applied to clocked devices
- Basic clock generation for test vectoring and failure analysis
- Digital triggering

The pulse generator can be purchased as an upgrade to existing systems (KTEI version 6.0 or above required) or as an option for new systems.

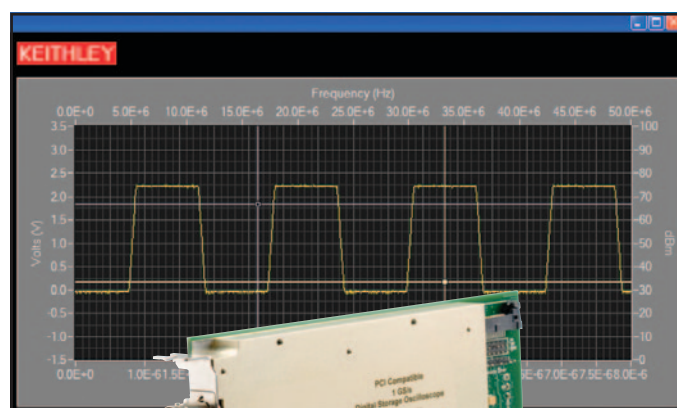
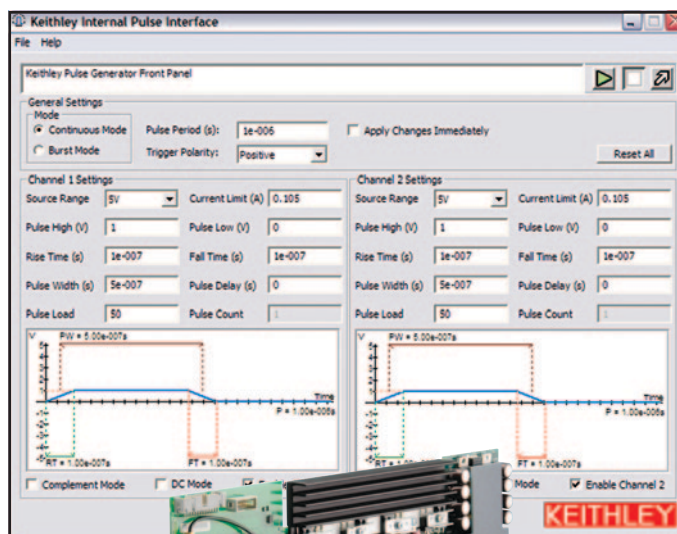
Dual-Channel Digital Oscilloscope

The optional dual-channel digital oscilloscopes place more than the performance of a bench-top oscilloscope into your 4200-SCS. They also support time-domain measurements of pulse waveforms and monitor the reactions of devices under test to those pulses. Some of the features of these oscilloscopes include: a broad selection of acquisition modes, triggers, measurements, calculations, and up to four reference waveforms.

The dual-channel oscilloscopes integrate directly into the Model 4200-SCS chassis. Either can be purchased as an upgrade to existing systems (KTEI version 6.0 or above required) or as an option for new systems.

Key Pulse Generator SPECIFICATIONS

Frequency Range	1Hz–50MHz
Pulse Width	Programmable from 10ns to near DC
Channels	Dual independent channels
Pulse Amplitude Range	100mV–20V into 50 Ω , 100mV–40V into 1M Ω
Programmable Parameters	Pulse width, duty cycle, rise time, fall time, amplitude, offset



Key Oscilloscope SPECIFICATIONS

	4200-SCP2	4200-SCP2HR (High Resolution)
Bandwidth	DC to 750MHz	DC to 250MHz
Channels	2	2
Maximum Sample Rate	1.25GS/s per channel	200MS/s per channel

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Semiconductor Characterization System DC I-V, C-V, and Pulse in One Test Environment

SERVICES AVAILABLE

4200-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4200-3Y-CAL	3 (Z540-1 compliant) calibrations within 3 years of purchase for Model 4200-SCS*
4200-FLASH-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4200-FLASH-3Y-CAL	3 (Z540-1 compliant) calibrations within 3 years of purchase for Model 4200-FLASH*
4200-PA-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4200-PIV-A-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4200-PIV-A-3Y-CAL	3 (Z540-1 compliant) calibrations within 3 years of purchase for Model 4200-PIV-A*
4200-PIV-Q-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4200-PIV-Q-3Y-CAL	3 (Z540-1 compliant) calibrations within 3 years of purchase for Model 4200-PIV-Q*
4200-SCP2-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4200-SCP2-3Y-CAL	3 (Z540-1 compliant) calibrations within 3 years of purchase for Model 4200-SCP2*
4205-PG2-3Y-EW	1-year factory warranty extended to 3 years from date of shipment
4205-PG2-3Y-CAL	3 (Z540-1 compliant) calibrations within 3 years of purchase for Model 4205-PG2*
IMPL-4200	1-day on-site implementation of TRN-4200-1-C
TRN-4200-1-C	Course: Optimizing the 4200-SCS for Your Application

*Not available in all countries

Application Packages

Optional application packages combine specific sets of hardware, interconnect, and Keithley developed code. They are described in the following pages.

Application packages designed for specific needs

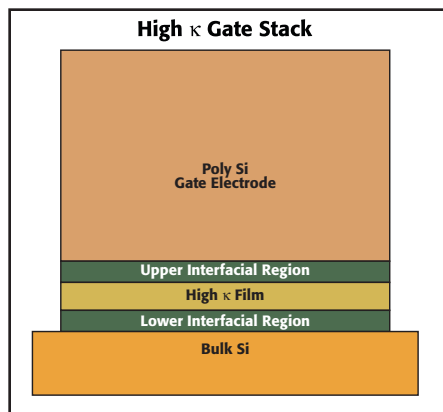
	4200-PIV-A	4200-PIV-Q	4200-FLASH
Description	For charge trapping and isothermal testing in lower technologies such as CMOS	For higher power pulse testing in III-V, LDMOS, and other higher frequency and higher power FET devices	For testing FLASH memory devices (NOR and NAND, including MLC technologies)
Device	FET	HEMT, FET	Floating gate FET
Technology	Advanced CMOS	III-V/LDMOS	NAND, NOR, nonvolatile memory
Source Method	Pulse gate, DC bias on drain	Dual pulse for gate and drain with quiescent point testing	Pulse gate, drain, source, and substrate
Measure Method	Pulse I-V and DC	Pulse I-V and DC	DC only
Measurements	Gate voltage, Drain voltage and current	Gate voltage and current, Drain voltage and current	Gate voltage and current, Drain voltage and current
Pulse Width Range¹	40ns to 150ns	500ns to 999ms	250ns to 1s
Unique Capability	8-bit, 1 gigasample/s measure rate, good for advanced CMOS Pulse I-V testing and high speed single-pulse charge trapping	Dual-channel, quiescent point pulsing for scaled-down RF transistors	One multi-level pulse channel per DUT pin, integrated High Endurance Output Relay supports endurance testing of NAND and NOR

1. Full Width Half Maximum (FWHM)

4200-SCS

Semiconductor Characterization System

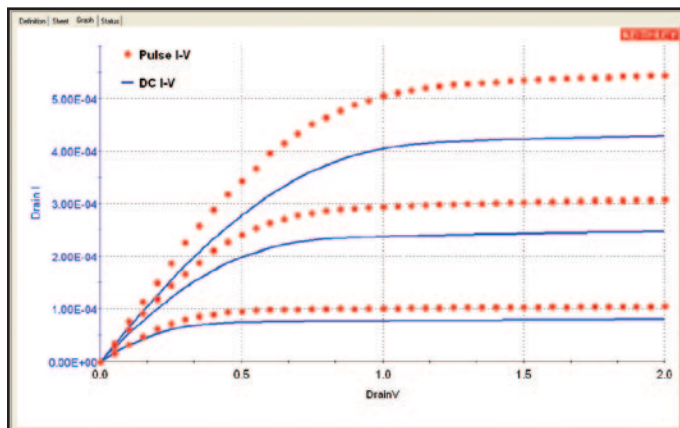
DC I-V, C-V, and Pulse in One Test Environment



Pulse I-V measurement capabilities are increasingly critical for high κ gate stack characterization and isothermal testing of new devices.



To minimize the signal reflections due to poor impedance matching that often plague "do-it-yourself" pulse testing systems, Keithley's Pulse I-V package includes a system interconnect setup that provides AC/DC coupling to connect the pulse generator and the DC instrumentation.



Pulse testing can characterize a device with little to no isothermal degradation.

4200-PIV-A: Pulse I-V Solution Package

The 4200-PIV-A Pulse I-V package provides a turnkey pulse I-V solution. It is a comprehensive package of hardware and software, designed to integrate seamlessly with the Model 4200-SCS workstation. It combines the dual-channel pulse generator, dual-channel digital oscilloscope, specialized interconnect, and patented Pulse I-V software.

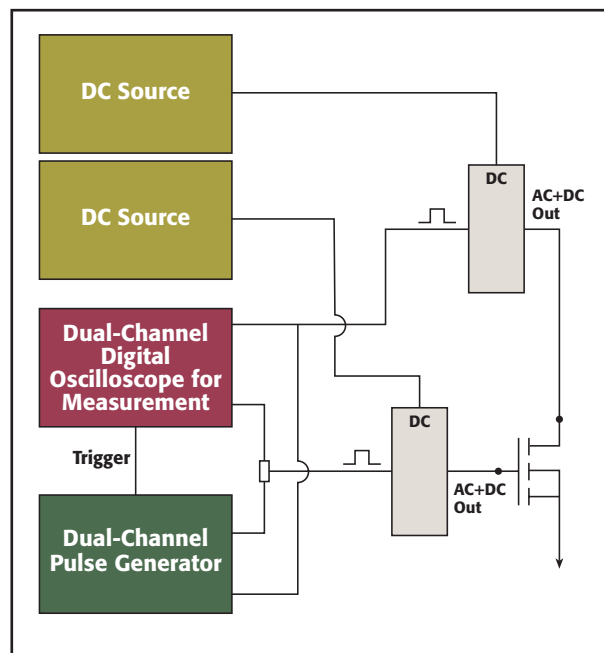
The Pulse I-V software controls sourcing (from the pulse generator) and data acquisition (from the oscilloscope) to automate a variety of Pulse I-V tests. Running in the Model 4200-SCS's proven interface, the Pulse I-V software provides instrument setup and control, data storage, and presentation. The innovative software includes both cable compensation and a solution to the load-line effect, producing pulsed-based I-V transistor curves, such as the V_{DS} - I_D family of curves and V_{GS} - I_D for voltage threshold extraction.

The Pulse I-V bundle allows the Model 4200-SCS to support a wide range of applications, such as charge trapping for high κ dielectric characterization, isothermal testing of devices and materials subject to self-heating effects, charge pumping, AC stress testing, clock generation, and mixed signal device testing.

The specialized interconnect solves most of the problems encountered in high speed pulse testing, such as:

- Combining pulse and DC sources to a single DUT pin to permit both DC and pulse characterization without recabling or switching
- Impedance matching for pulse integrity to minimize reflection
- Straightforward cabling and connection to the DUT for easy setup

NEW! 4200-MMPC-X Multi-Measurement Cable Set allows easy changeover from I-V to C-V to PIV



The Pulse I-V package includes everything needed to implement a turnkey system for pulsed I-V testing of leading-edge devices and materials. Pieces included in the package are:

- Integrated dual-channel pulse generator
- Dual-channel digital oscilloscope
- Pulse I-V control software (patent pending)
- Interconnect designed to minimize the signal reflections common to pulse I-V testing (patent pending)
- All required connectors and cables
- Sample projects for:
 - Pulse I-V isothermal testing of FinFETs, SOI devices, and other devices with self-heating problems
 - Charge-trap testing for high κ gate stack characterization

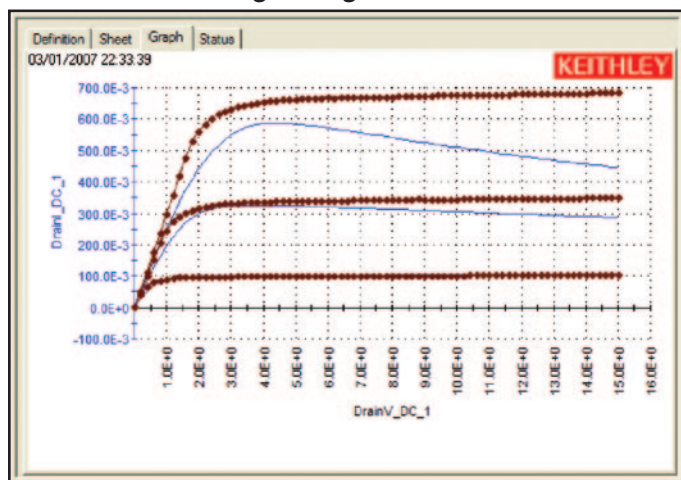
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4200-PIV-Q: Pulsed I-V, Q Point, Dual-Channel, Pulsing Package



The 4200-PIV-Q package is designed for quiescent point pulsing of scaled-down RF transistors, such as HEMT and FET devices in III-V or LDMOS technologies. It can be used for a variety of large signal tests on high frequency transistors as well as for investigation of dispersion phenomena and device performance at speed.

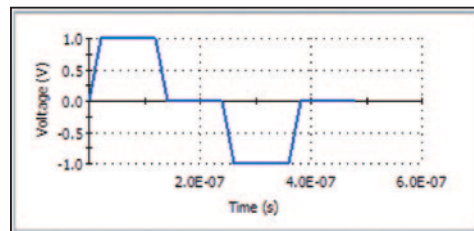
This package includes multiple 4205-PG2 pulse generators and the 4200-SCP2HR oscilloscope and offers capabilities such as dual-channel pulsing (for pulsing on both the gate and the drain simultaneously), higher power pulsing than the 4200-PIV-A package, and pulsing from a non-zero quiescent point. Some of its features include:

- Elimination or characterization of thermal issues
 - Ability to compare DC vs. Pulse for dispersion effects
 - Software and interconnect for Quiescent point testing
 - Test code for typical characterization tests
 - Pulse widths adjustable from 500ns to near-DC (999ms)
 - Ability to use the same setup for performing true DC tests without re-cabling the system
 - Dual-channel pulse I-V testing for III-V and LDMOS:
 - Pulse voltage on gate and drain
 - Measure gate current, drain voltage, and current
 - $\pm 20V$ pulses for the gate, $\pm 38V$ pulses for the drain
- Some of the specific tests are:
- $V_{DS}-I_D$: Both pulse and DC
 - $V_{GS}-I_D$: Both pulse and DC
 - Single-pulse scope view, which is useful for setup validation, pulse width optimization, and prototyping of novel pulse tests

4200-FLASH: Non-Volatile Memory Test Package

	Start (V)	Stop (V)	Time (s)	Trig (1/0)	SSR (1/0)
1	0	1	2.00E-08	1	1
2	1	1	1.00E-07	1	1
3	1	0	2.00E-08	0	0
4	0	0	1.00E-07	0	0

Easy setup for program/erase cycles



Typical NOR FLASH gate program/erase cycle

The optional 4200-FLASH application package tests single FLASH memory cells or small arrays quickly and easily using four (or up to eight optional) independent, but synchronized, multi-level pulse channels. It includes all the code and interconnect needed to perform a standard set of FLASH memory tests for NAND and NOR technologies, such as characterization, endurance, and disturb tests. It also supplies the higher pulse voltages that are important for MLC technologies.

4200-FLASH takes advantage of the new patent-pending Segment ARB™ waveform generator, which makes typical FLASH program/erase cycles simple to set up and run on a single pulse channel. It also combines the Segment ARB waveform generator with the in-line high endurance relay for Endurance tests. The tight control of this output relay can shorten lifetime test times significantly.

4200-FLASH provides four (or up to eight optional) channels of multi-level pulse that support:

- $\pm 40V$ pulsing into a high impedance pin ($\pm 20V$ into 50Ω)
- High endurance output relay which provides fast open/close for pin isolation during an erase pulse
- Pulse widths from 200ns to 1s
- Up to 25 pulse levels (100 pulse segments)

NEW! 4200-MMPC-X Multi-Measurement Cable Set allows easy changeover from I-V to C-V to PIV

Other 4200-FLASH features include:

- Code for performing tests on floating gate FLASH and embedded NVM memory
- Performs linear or log-based DC measurements for Disturb and Endurance tests based on the number of program/erase cycles
- Controls switching between program/erase and DC characterization without using a switch matrix
- Full support for multi-level cell technology with up to $\pm 40V$ pulsing on the gate
- Solid state relays with high endurance output relay (HEOR) capability for pin disconnect within Program+Erase waveform

4200-SCS

Semiconductor Characterization System

DC I-V, C-V, and Pulse in One Test Environment

OPTIONAL INSTRUMENTATION AND ACCESSORIES

OPTIONAL INSTRUMENTATION

4210-CVU	Integrated C-V Instrument
4200-SMU	Medium Power Source-Measure Unit for 4200-SCS. 100mA to 100fA, 200V to 1μV, 2 Watt
4210-SMU	High Power Source-Measure Unit for 4200-SCS. 1A to 100fA, 200V to 1μV, 20 Watt
4200-PA	Remote PreAmp Option for 4200-SMU and 4210-SMU, extends SMU to 0.1fA resolution
4205-PG2	Dual-Channel Pulse Generator
4200-SCP2	Dual-Channel Integrated Oscilloscope
4200-SCP2HR	High Resolution, Dual-Channel Integrated Oscilloscope
4200-SCP2-ACC	Optional Scope Probe

OPTIONAL APPLICATION PACKAGES

4200-PIV-A	Complete Pulse I-V Package for leading edge CMOS
4200-PIV-Q	Pulse I-V Package with Q point and dual-channel pulsing
4200-FLASH	Non-volatile Memory Test Package

OPTIONAL SWITCHING SYSTEMS AND CARDS

SYSTEMS

707A	6-Slot Switching Matrix Mainframe
708A	Single-Slot Switching Matrix Mainframe

CARDS

7071	8×12, General Purpose, Matrix Card
7071-4	Dual 4×12, General Purpose, Matrix Card
7072	8×12, Semiconductor Matrix Card
7072-HV	8×12, High Voltage, Semiconductor Matrix Card
7075	Eight 1×12, Two-Pole, Multiplexer Card
7173-50	4×12, Two-Pole, High Frequency, Matrix Card
7174A	8×12, High Speed, Low Leakage Current, Matrix Card

OPTIONAL ACCESSORIES

CONNECTORS AND ADAPTERS

CS-565	Female BNC to Female BNC Adapter
CS-701	BNC Tee Adapter (female, male, female)
CS-719	3-lug Triax Jack Receptacle
CS-1247	SMA Female to BNC Male Adapter
CS-1249	SMA Female to SMB Plug Adapter
CS-1251	BNC Female to SMB Plug Adapter
CS-1252	SMA Male to BNC Female Adapter
CS-1281	SMA Female to SMA Female Adapter
CS-1382	Female MMBX Jack to Male SMA Plug Adapter
CS-1390	Male LEMO Triax to Female SMA Adapter
CS-1391	SMA Tee Adapter (female, male, female)
CS-1479	SMA Male to BNC Male Adapter

237-BAN-3A	Triax Cable Center Conductor terminated in a safety banana plug
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237-BNC-TRX	Male BNC to 3-lug Female Triax Adapter
237-TRX-BAR	3-lug Triax Barrel Adapter (female to female)
237-TRX-T	3-slot Male to Dual 3-lug Female Triax Tee Adapter

7078-TRX-BNC	3-Slot Male Triax to BNC Adapter
7078-TRX-GND	3-Slot Male Triax to Female BNC Connector (guards removed)

CABLES AND CABLE SETS

NOTE: All 4200-SCS systems and instrument options are supplied with required cables, 2m (6.5 ft) length.

CA-19-2	BNC to BNC Cable, 1.5m
CA-404B	SMA to SMA Coaxial Cable, 2m
CA-405B	SMA to SMA Coaxial Cable, 15cm
CA-406B	SMA to SMA Coaxial Cable, 33cm
CA-446A	SMA to SMA Coaxial Cable, 3m
CA-447A	SMA to SMA Coaxial Cable, 1.5m
CA-451A	SMA to SMA Coaxial Cable, 10.8cm
CA-452A	SMA to SMA Coaxial Cable, 20.4cm
236-ILC-3	Safety Interlock Cable, 3m
237-ALG-2	Low Noise Triax Input Cable terminated with 3 alligator clips, 2m
4210-MMPC-C	Multi-Measurement (I-V, C-V, Pulse) Prober Cable Kit for Cascade Microtech 12000 prober series
4210-MMPC-S	Multi-Measurement (I-V, C-V, Pulse) Prober Cable Kit for SUSS MicroTec PA200/300 prober series
4200-MTRX-*	Ultra Low Noise SMU Triax Cable: 1m, 2m, and 3m options
4200-PRB-C	SMA to SSMC Y Cable with local ground
4200-RPC-*	Remote PreAmp Cable: 0.3m, 2m, 3m, 6m options
4200-TRX-*	Ultra Low Noise PreAmp Triax Cable: 0.3m, 2m, 3m options
7007-1	Double-Shielded Premium GPIB Cable, 1m
7007-2	Double-Shielded Premium GPIB Cable, 2m

FIXTURES

8101-4TRX	4-Pin Transistor Fixture
8101-PIV	Pulse I-V Demo Fixture
LR8028	Component Test Fixture

CABINET MOUNTING ACCESSORIES

4200-RM	Fixed Cabinet Mount Kit
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REMOTE PREAMP MOUNTING ACCESSORIES

4200-MAG-BASE	Magnetic Base for mounting 4200-PA on a probe platen
4200-TMB	Triaxial Mounting Bracket for mounting 4200-PA on a triaxial mounting panel
4200-VAC-BASE	Vacuum Base for mounting 4200-PA on a prober platen

COMPUTER ACCESSORIES

4200-MOUSE	Microsoft Ambidextrous 2 Button Mouse (Note: A pointing device is integrated with the 4200-SCS keyboard.)
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SOFTWARE

ACS-BASIC	Component Characterization Software
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DRIVERS

4200ICCAP-6.0	IC-CAP Driver and Source Code for 4200-SCS: UNIX/Windows
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OTHER ACCESSORIES

EM-50A	Modified Power Splitter
TL-24	SMA Torque Wrench
4200-CART	Roll-Around Cart for 4200-SCS
4200-CASE	Transport Case for 4200-SCS
4200-MAN	Printed Manual Set

ADAPTER, CABLE, AND STABILIZER KITS

4200-CVU-PWR	CVU Power Package for ±200V C-V
4200-CVU-PROBER-KIT	Accessory Kit for connection to popular analytical probes
4200-Q-STBL-KIT	Addresses oscillation when performing pulse I-V tests on RF transistors

SUPPLIED ACCESSORIES

ACCESSORIES SUPPLIED WITH EACH MODEL 4210-CVU:

CA-447A	SMA Cables, male to male, 100Ω, 1.5m (5 ft.) (4)
CS-1247	Female SMA to Male BNC Adapters (4)
CS-701	BNC Tee Adapters (2)
TL-24	SMA Torque Wrench

ACCESSORIES SUPPLIED WITH EACH MODEL 4200-SMU OR 4210-SMU:

4200-MTRX-2	Two Ultra Low Noise SMU Triax Cables, 2m (6.6 ft). Not included with SMUs configured with a 4200-PA Remote PreAmp.
4200-TRX-2	Ultra Low Noise PreAmp Triax Cable, 2m (6.6 ft). Two supplied for Ground Unit. Two supplied in replacement of 4200-MTRX-2 cables for each SMU configured with a 4200-PA.
4200-RPC-2	Remote PreAmp Cable, 2m (6.6 ft). One supplied for each PreAmp.
236-ILC-3	Interlock Cable, 3m (10 ft)
Line Cord	NEMA 5-15P for 100-115VAC or CEE 7/7 (Continental European) for 240VAC

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DC SMU CURRENT SPECIFICATIONS

		CURRENT RANGE ¹	MAX. VOLTAGE	MEASURE		SOURCE	
				Resolution ³	Accuracy ±(% rdg + amps)	Resolution ³	Accuracy ±(% rdg + amps)
4210-SMU ² High Power SMU	4200-SMU ² Medium Power SMU	1 A	21 V	1 μA	0.100% + 200 μA	50 μA	0.100% + 350 μA
		100 mA	210 V	100 nA	0.045% + 3 μA	5 μA	0.050% + 15 μA
		100 mA	21 V	100 nA	0.045% + 3 μA	5 μA	0.050% + 15 μA
	4200-SMU ² Medium Power SMU	10 mA	210 V	10 nA	0.037% + 300 nA	500 nA	0.042% + 1.5 μA
		1 mA	210 V	1 nA	0.035% + 30 nA	50 nA	0.040% + 150 nA
		100 μA	210 V	100 pA	0.033% + 3 nA	5 nA	0.038% + 15 nA
		10 μA	210 V	10 pA	0.050% + 600 pA	500 pA	0.060% + 1.5 nA
	4200-SMU and 4210-SMU with optional 4200-PA PreAmp	1 μA	210 V	1 pA	0.050% + 100 pA	50 pA	0.060% + 200 pA
		100 nA	210 V	100 fA	0.050% + 30 pA	5 pA	0.060% + 30 pA
		10 nA	210 V	10 fA	0.050% + 1 pA	500 fA	0.060% + 3 pA
4200-SMU and 4210-SMU with optional 4200-PA PreAmp	1 nA	210 V	3 fA	0.050% + 100 fA	50 fA	0.060% + 300 fA	
	100 pA	210 V	1 fA	0.100% + 30 fA	15 fA	0.100% + 80 fA	
	10 pA	210 V	0.3 fA	0.500% + 15 fA	5 fA	0.500% + 50 fA	
	1 pA	210 V	100 aA	1.000% + 10 fA	1.5 fA	1.000% + 40 fA	

VOLTAGE COMPLIANCE: Bipolar limits set with a single value between full scale and 10% of selected voltage range.

DC SMU VOLTAGE SPECIFICATIONS

VOLTAGE RANGE ¹	MAX. CURRENT		MEASURE		SOURCE	
	4200-SMU	4210-SMU	Resolution ³	Accuracy ±(% rdg + volts)	Resolution ³	Accuracy ±(% rdg + volts)
200 V ⁴	10.5 mA	105 mA	200 μ V	0.015% + 3 mV	5 mV	0.02% + 15 mV
20 V	105 mA	1.05 A	20 μ V	0.01 % + 1 mV	500 μ V	0.02% + 1.5 mV
2 V	105 mA	1.05 A	2 μ V	0.012% + 150 μ V	50 μ V	0.02% + 300 μ V
200 mV	105 mA	1.05 A	1 μ V	0.012% + 100 μ V	5 μ V	0.02% + 150 μ V

CURRENT COMPLIANCE: Bipolar limits set with a single value between full scale and 10% of selected current range.

Supplemental DC SMU Information

Supplemental information is not warranted but provides useful information about the Models 4200-SMU, 4210-SMU, and 4200-PA.

COMPLIANCE ACCURACY:

Voltage compliance equals the voltage source specifications.
Current compliance equals the current source specifications.

OVERSHOOT: <0.1% typical.

Voltage: Full scale step, resistive load, and 10mA range.
Current: 1mA step, $R_L = 10k\Omega$, 20V range.

RANGE CHANGE TRANSIENT:

Voltage Ranging: <200mV.
Current Ranging: <200mV.

ACCURACY SPECIFICATIONS: Accuracy specifications are multiplied by one of the following factors, depending upon the ambient temperature and humidity.

Temperature	% Relative Humidity	
	5–60	60–80
10°–18°C	×3	×3
18°–28°C	×1	×3
28°–40°C	×3	×5

REMOTE SENSE: <10 Ω in series with FORCE terminal not to exceed a 5V difference between FORCE and SENSE terminals.
 $\pm 30V$ maximum between COMMON and SENSE LO.

MAXIMUM LOAD CAPACITANCE: 10nF.

MAXIMUM GUARD OFFSET VOLTAGE: 3mV from FORCE.

GUARD OUTPUT IMPEDANCE: 100k Ω .

MAXIMUM GUARD CAPACITANCE: 1500pF.

MAXIMUM SHIELD CAPACITANCE: 3300pF.

4200-SMU and 4210-SMU SHUNT RESISTANCE (FORCE to COMMON): >10¹² Ω (100nA–1 μ A ranges).

4200-PA SHUNT RESISTANCE (FORCE to COMMON): >10¹⁶ Ω (1pA and 10pA ranges), >10¹³ Ω (100pA–100nA ranges).

OUTPUT TERMINAL CONNECTION: Dual triaxial connectors for 4200-PA, dual mini-triaxial connectors for 4200-SMU and 4210-SMU.

NOISE CHARACTERISTICS (typical):

Voltage Source (rms): 0.01% of output range.
Current Source (rms): 0.1% of output range.
Voltage Measure (p-p): 0.02% of measurement range.
Current Measure (p-p): 0.2% of measurement range.

MAXIMUM SLEW RATE: 0.2V/ μ s.

SPECIFICATION CONDITIONS

Specifications are the performance standards against which the Models 4200-SMU, 4210-SMU, and 4200-PA are tested. The measurement and source accuracy are specified at the termination of the supplied cables.

- 23°C $\pm 5^\circ$ C, within 1 year of calibration, RH between 5% and 60%, after 30 minutes of warm-up.
- Speed set to NORMAL.
- Guarded Kelvin connection.
- $\pm 1^\circ$ C and 24 hours from ACAL.

NOTES

1. All ranges extend to 105% of full scale.
2. Specifications apply on these ranges with or without a 4200-PA.
3. Specified resolution is limited by fundamental noise limits. Measured resolution is 6 $\frac{1}{2}$ digits on each range. Source resolution is 4 $\frac{1}{2}$ digits on each range.
4. Interlock must be engaged to use the 200V range.

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Additional DC SMU Specifications

MAX. OUTPUT POWER: 22 watts for 4210-SMU and 2.2 watts for 4200-SMU (both are four-quadrant source/sink operation).

DC FLOATING VOLTAGE: COMMON can be floated ± 32 volts from chassis ground.

VOLTAGE MONITOR (SMU in VMU mode)

Voltage Range	Measure Resolution	Measure Accuracy $\pm(\%rdg + \text{volts})$
200 V	200 μV	0.015% + 3 mV
20 V	20 μV	0.01% + 1 mV
2 V	2 μV	0.012% + 110 μV
200 mV	1 μV	0.012% + 80 μV

INPUT IMPEDANCE: $>10^{13}\Omega$.

INPUT LEAKAGE CURRENT: $<30\text{pA}$.

MEASUREMENT NOISE: 0.02% of measurement range (rms).

DIFFERENTIAL VOLTAGE MONITOR

Differential Voltage Monitor is available by measuring with two SMUs in VMU mode or by using the low sense terminal provided with each SMU.

GROUND UNIT

Voltage error when using the ground unit is included in the 4200-SMU, 4210-SMU, and 4200-PA specifications. No additional errors are introduced when using the ground unit.

OUTPUT TERMINAL CONNECTION: Dual triaxial, 5-way binding post.

MAXIMUM CURRENT: 2.6A using dual triaxial connection; 8.5A using 5-way binding posts.

LOAD CAPACITANCE: No limit.

CABLE RESISTANCE: FORCE $\leq 1\Omega$, SENSE $\leq 10\Omega$.

GENERAL

TEMPERATURE RANGE

Operating: $+10^\circ$ to $+40^\circ\text{C}$.

Storage: -15° to $+60^\circ\text{C}$.

HUMIDITY RANGE

Operating: 5% to 80% RH, non-condensing.

Storage: 5% to 90% RH, non-condensing.

ALTITUDE

Operating: 0 to 2000m.

Storage: 0 to 4600m.

POWER REQUIREMENTS: 100V to 240V, 50 to 60Hz.

MAXIMUM VA: 1000VA.

REGULATORY COMPLIANCE:

Safety: Low Voltage Directive 73/23/EEC.

EMC: Directive 89/336/EEC.

DIMENSIONS: 43.6cm wide $\times$ 22.3cm high $\times$ 56.5cm deep (17 $\frac{3}{8}$ in $\times$ 8 $\frac{3}{4}$ in $\times$ 22 $\frac{1}{4}$ in).

WEIGHT (approx.): 29.7kg (65.5 lbs) for typical configuration of four SMUs.

I/O PORTS: USB, SVGA, Printer, RS-232, GPIB, Ethernet, Mouse, Keyboard.

4205-PG2 Dual-Channel Pulse Generator Specifications^{1, 2}

PULSE/LEVEL³

		High Speed	High Voltage
V_{OUT}	50 Ω into 50 Ω	-5V to +5V	-20V to +20V
V_{OUT}	50 Ω into 1 M Ω	-10V to +10V	-40V to +40V
Accuracy		$\pm(3\% + 50 \text{ mV})$	$\pm(3\% + 100 \text{ mV})$
Amplitude/Level	50 Ω into 50 Ω	1 mV	5 mV
Resolution	50 Ω into 1 M Ω	2 mV	10 mV
Output Connectors		SMA	SMA
Source Impedance		50 Ω Nominal	50 Ω Nominal
Accuracy		1%	1%
Short Circuit Current		$\pm 200 \text{ mA}$	$\pm 800 \text{ mA}$
Current into 50 Ω Load (at full scale)		$\pm 100 \text{ mA}$ typical	$\pm 400 \text{ mA}$ typical
Baseline Noise		$\pm(0.1\% + 5 \text{ mV})$ RMS typical	$\pm(0.1\% + 5 \text{ mV})$ RMS typical
Overshoot/Pre-shoot/Ringing		$\pm 5\%$ of amplitude $\pm 20 \text{ mV}$	$\pm 5\%$ of amplitude $\pm 80 \text{ mV}$
Output Limit		Programmable limit to protect the DUT	

TIMING

	High Speed	High Voltage
Frequency Range	1 Hz to 50 MHz	1 Hz to 2 MHz
Timing Resolution	10 ns	10 ns
RMS Jitter (period, width)	0.01 % + 200 ps typical	0.01 % + 200 ps typical
Period Range	20 ns to 1 s	500 ns to 1 s
Accuracy	$\pm 1\%$	$\pm 1\%$
Pulse Width Range	10ns to (period - 10ns)	250ns to (period - 100ns)
Accuracy	$\pm(3\% + 200 \text{ ps})$	$\pm(3\% + 5 \text{ ns})$
Programmable	10 ns-33 ms	100 ns-33 ms
Transition Time (0-100%)		
	Accuracy	$\pm 1\%$ for transition time
		$<100 \text{ ns}$
Transition Slew Rate ⁴		$<1 \mu\text{s}$
	Linearity	3% for transition time
		$<100 \text{ ns}$
		$<15 \text{ ns}$
Typical Minimum Transition Time 10-90%	Pulse Period and width are variable in 10 ns steps without any output glitches or dropouts	
Solid State Relay	Open or close time	100 μs

TRIGGER

TRIGGER OUTPUT IMPEDANCE: 50 Ω .

TRIGGER OUTPUT LEVEL: TTL.

TRIGGER IN IMPEDANCE: 10k Ω .

TRIGGER IN LEVEL: TTL.

TRIGGER IN TRANSITION TIMING, MAXIMUM: $<100\text{ns}$.

TRIGGER IN TO PULSE OUT DELAY: 560ns.

TRIGGER SYNCHRONIZATION/JITTER⁵: $<8\text{ns}$.

NOTES

- Unless stated otherwise, all specifications assume a 50 Ω termination.
- Maximum number of PG2 cards in the 4200 chassis is 4.
- Level specifications are valid after 50ns typical settling time (after slewing) for the high speed mode and after 500ns typical settling time (after slewing) for the high voltage mode into a 50 Ω load.
- Specifications apply to a 10-90% transition, typical. Minimum slew rate for high speed range = 724mV/ms. For high voltage range = 2.71V/ms, which applies to both the standard pulse and Segment ARB™ mode.
- For multiple 4205-PG2 cards, when using appropriate cabling and the "trigger per waveform" trigger mode.

All specifications apply at $23^\circ \pm 5^\circ\text{C}$, within one year of calibration, RH between 5% and 60%, after 30 minutes of warmup.

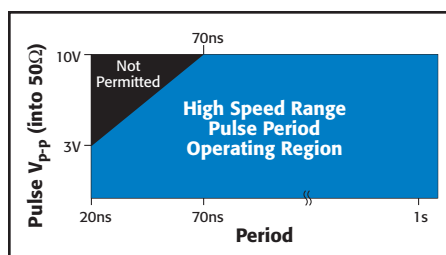


Figure 1. Permitted area of operation.

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4200-SCP2 1.25GS Dual-Channel Oscilloscope Card and 4200-SCP2HR 200MS Dual-Channel Oscilloscope Card Specifications¹

ANALOG INPUT¹

	4200-SCP2	4200-SCP2HR
No. of Channels	2	2
Bandwidth (50Ω)	DC to 750 MHz	DC to 250 MHz, typical
Bandwidth (1MΩ)	DC to 350 MHz	DC to 125 MHz, typical
Full Scale Input Range (50 Ω)	0.05, 0.1, 0.25, 0.5, 1, 2, 5, 10 (Vp-p)	0.05, 0.1, 0.25, 0.5, 1, 2, 5, 10 (Vp-p)
Full Scale Input Range (1 MΩ)	0.1, 0.2, 0.5, 1, 2.5, 5, 10, 20, 50, 100 (Vp-p)	0.25, 0.5, 1.25, 2.5, 5, 10, 25, 50 (Vp-p)
DC Gain Accuracy	<±1% of full scale	< ±0.25% of full scale
Impedance	1 MΩ 12 pF or 50 Ω	1 MΩ 12 pF or 50 Ω
Impedance Accuracy	±1%	±1%
Coupling	DC or AC	DC or AC
Offset Adjust	±(full scale range/2)	±(full scale range/2)
Offset Accuracy	±(1% offset + 1% full scale)	±1%
Input Connectors	BNC	BNC
Absolute Maximum Input (50 Ω)	±5V DC	±5V DC
Absolute Maximum Input (1 MΩ)	±210V DC	±210V DC

ANALOG-TO-DIGITAL CONVERTER

	4200-SCP2	4200-SCP2HR
Resolution	8 bit	16 bit
Sample Rate	2.5 kS/s to 1.25 GS/s in 1, 2.5, 5 steps	10 kS/s to 200 MS/s in 1, 2.5, 4, 5 steps
Memory Depth	2.5 GS/s (1 channel interleaved) 1 MS/channel	400 MS/s (1 channel interleaved) 1 MS/channel
Acquisition Time Range	2 MS on 1 channel, interleaved 50 ns to 419 seconds	2 MS on 1 channel, interleaved 250 ns to 3,355 seconds
Acquisition Modes	Normal, Average, Envelope, and Equivalent-time	Normal, Average, Envelope, and Equivalent-time

TRIGGER

	4200-SCP2	4200-SCP2HR
Trigger Source	Channels 1 or 2, External, Pattern, Software	Channels 1 or 2, External, Pattern, Software
Post-Trigger Delay	0 to 655 seconds	0 to 655 seconds
Pre-Trigger Delay	0 to waveform time	0 to waveform time
Trigger Hold Off Range	0 to 655 seconds	0 to 655 seconds
Trigger Modes	Edge or Pulse Width	Edge or Pulse Width
Edge Trigger Mode	Rising or Falling Edge	Rising or Falling Edge
Pulse Width Range	20ns to 655 seconds, 10ns resolution	20ns to 655 seconds, 10ns resolution
External Trigger Input	TTL Compatible, 10 kΩ input impedance	TTL Compatible, 10 kΩ input impedance
Connector	SMB	SMB

OPTIONAL SCOPE PROBE: 4200-SCP2-ACC

BANDWIDTH: 70MHz (4200-SCP2); 15MHz (4200-SCP2HR).

ATTENUATION: 1×.

MAX DC: 300V DC rated.

LOADING: 100pF and 1MΩ.

LENGTH: 1m.

CONNECTOR: BNC.

NOTES

1. Inputs are referenced to 4200 chassis ground
All specifications apply at 23±5°C, within 1 year of calibration, RH between 5% and 60%, after 30 minutes of warmup.

4200-PIV-A Pulse I-V Option Specifications¹

CHANNELS: 2.

TYPICAL PULSE PERFORMANCE (with 4205 Remote Bias Tee⁴):

Measurement Accuracy: <4% of signal ±1mV.

Maximum Current Measure: 100mA.

Resolution: 100nA².

Offset: <500nA.

Sample Rate: 1GS/s.

Duty Cycle: <0.1%.

DC Offset: ±200V.

Minimum Transition Time (10–90%): <15ns.

Pulse Source Voltage Range: 0 to ±5V into gate.

Pulse Width: 40ns to 150ns.

SMU TYPICAL DC PERFORMANCE (with 4205 Remote Bias Tee):

Leakage: 1–10nA/V³.

Noise: 1–10nA RMS.

Maximum Voltage: 210V (>40V requires safety interlock and related precautions).

Maximum Current: 0.5A.

4200 REMOTE BIAS TEE TYPICAL PERFORMANCE:

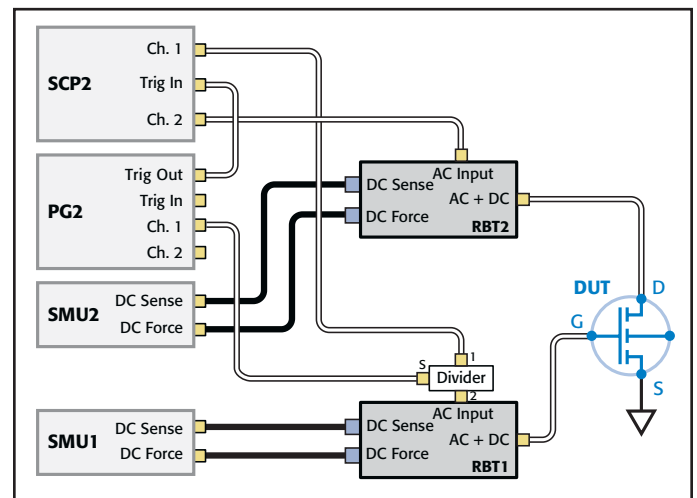
Band Pass: 3.5kHz–300MHz (3dB).

Power Divider Max Power Input: 0.125W DC.

NOTES

1. Unless stated otherwise, all specifications assume a 50 termination.
2. When using Adaptive filtering.
3. Leakage measured after a 5 second settling time.
4. All typical specs apply to the AC+DC output connector of the 4205 Remote Bias Tee and after system compensation.

All specifications apply at 23±5°C, within one year of calibration, RH between 5% and 60%, after 30 minutes of warmup.



Interconnection for 4200-PIV-A for leading edge CMOS, high κ , and isothermal testing. PIV-A pulses the voltage on the gate and provides a DC bias on the drain.

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4200-PIV-Q Typical Specifications¹

CHANNELS: 2.

TYPICAL PULSE PERFORMANCE⁴:

Measurement Accuracy: Gate Current: $<50\mu\text{A}$ offset, $10\mu\text{A}$ resolution².
Drain Current: $<100\mu\text{A}$ offset, $10\mu\text{A}$ resolution².
Maximum Current Measure: Gate: 100mA (into 50Ω)⁶.
Drain: 760mA (into 50Ω), 1.33A into 5Ω ⁶.

Sample Rate: 200MS/s .

Duty Cycle: 0.001% to 99.9% .

Minimum Transition Time (10–90%): 150ns .

Gate Pulse Source: -20V to $+20\text{V}$.

Drain Voltage Range: -38V to $+38\text{V}$ (into 50Ω), $\pm 75\text{V}$ (into $1\text{k}\Omega$)⁶.

Pulse Width: 500ns to 999ms .

Pulse Period: 510ns to 1s .

SMU TYPICAL DC PERFORMANCE:

Typical DC Leakage, Gate: $<20\text{nA}$ offset for $<35\text{V}$.

Typical DC Leakage, Drain: $<10\text{nA/V}^3$ for $<35\text{V}$.

Typical DC Noise, Gate: $<20\text{nA RMS}$.

Gate Offset: $<20\text{nA}$.

Typical DC Noise, Drain: $<300\text{pA RMS}$.

Maximum Voltage: 210V ($>40\text{V}$ requires safety interlock and related precautions).

Maximum Current: 1A ⁵.

NOTES

1. Unless stated otherwise, all specifications assume a 50Ω termination.
2. Offset and resolution specified when using adaptive filtering after system cable compensation and offset correction.
3. Leakage measured after a 5 second settling time.
4. All typical specs apply to the AC+DC output cable (from the SMU Force, connected to the SMA tee attached to Triax to SMA adapter) after system compensation.
5. For the high power 4210-SMU. For the medium power 4200-SMU, the maximum current is 100mA .
6. Drain Pulse Source is a voltage pulser with 55Ω output impedance. To calculate the approximate maximum Drain current for any DUT resistance:
$$I_{\text{dmax}} = 80\text{V}/55 + R_{\text{DS}}$$

To calculate approximate maximum Drain voltage, input the I_{max} calculated above:
$$V_{\text{dmax}} = I_{\text{dmax}} \times R_{\text{DS}}$$

All specifications apply at $23^\circ \pm 5^\circ\text{C}$, within 1 year of calibration, RH between 5% and 60%, after 30 minutes of warmup.

4200-FLASH Typical Specifications¹

CHANNELS: 4 channels (optional 8 channels max.).

TYPICAL PULSE PERFORMANCE:

Number of Voltage Levels/Waveform: 25.

Minimum Transition Time: 150ns .

Pulse Source Voltage Range: 0 to $\pm 20\text{V}$ into 50Ω . 0 to $\pm 40\text{V}$ into high impedance.

Pulse Width: 250ns to 1s .

Trigger Synchronization/Jitter: $\pm 8\text{ns}$.

Switching Time for DUT Pin Isolation: $100\mu\text{s}$.

HEOR Off Capacitance: 250pF .

SMU TYPICAL DC PERFORMANCE

Typical DC Leakage: $<10\text{nA/V}^2$ for $<35\text{V}$.

Typical DC Noise: $<300\text{pA RMS}$.

Maximum Voltage: 200V ($>40\text{V}$ requires safety interlock and related precautions).

Maximum Current: 1A ³.

NOTES

1. Unless stated otherwise, all specifications assume a 50Ω termination
2. Leakage measured after a 5 second settling time
3. For the high power 4210-SMU. For the medium power 4200-SMU, the maximum current is 100mA

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